
Vocabulary Builder MVP

Phase 1 — Foundation (Web, Storage, Authentication)

Status:  Completed

Audience: Technical + Business

Region: `us-east-1`

1. Phase 1 Objective (Business + Technical)

Phase 1 establishes the **secure foundation** for the Vocabulary Builder MVP.

Business goals

- Deliver a **fast, global web experience** for students.
- Securely store uploaded learning documents.
- Prepare the system for AI-powered text processing in later phases.
- Separate responsibilities clearly to reduce risk and cost.

Technical goals

- Use AWS best practices (least privilege, private buckets, encryption).
 - Separate **website hosting** from **document processing**.
 - Use managed authentication (Cognito) instead of custom auth logic.
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2. High-Level Architecture

Core idea

We intentionally use **two separate S3 buckets**:

Bucket	Purpose
<code>vb-website-2025</code>	Static frontend (SPA)
<code>vb-documents-2025</code>	PDFs + AI outputs (Textract / Bedrock)

This separation:

- Improves security
- Simplifies IAM policies
- Prevents accidental data exposure

3. Architecture Diagram (Phase 1)

flowchart LR

U[User / Student]

CF[CloudFront
HTTPS + OAC]

WB[S3 Website Bucket
`vb-website-2025`
Private]

COG[Cognito Hosted UI
User Pool]

DB[S3 Documents Bucket
`vb-documents-2025`
Private]

U --> CF

CF --> WB

U --> COG

COG --> U

U --> DB

4. Components Breakdown

4.1 S3 Website Bucket (`vb-website-2025`)

Purpose: Host the frontend (HTML, JS, CSS)

Security controls

- Block Public Access: **ON**
- Object Ownership: **BucketOwnerEnforced**
- Encryption: **SSE-S3 (AES256)**

Access model

-  No public access
 -  Read-only access **only via CloudFront (OAC)**
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4.2 CloudFront Distribution

Purpose: Global CDN + HTTPS enforcement

Key settings

- Origin: S3 (**vb-website-2025**)
- Origin Access Control (OAC):
 - Signing protocol: SigV4
 - Signing behavior: Always
- Viewer protocol policy: **redirect-to-https**
- Default root object: **index.html**

Result

- Users cannot bypass CloudFront to access S3 directly
 - Website is fast and secure worldwide
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4.3 S3 Documents Bucket (**vb-documents-2025**)

Purpose

- Store uploaded PDFs
- Store AI processing outputs (future phases)

Security baseline

- Block Public Access: **ON**
- Encryption at rest: **SSE-S3**
- Versioning: **Enabled**
- ACLs disabled

Folder structure

vb-documents-2025/

|— raw/ # User uploaded PDFs
|— textract-output/ # Phase 2
|— comprehend-output/ # Phase 2

Lifecycle rules

Prefix	Expiration
raw/	90 days
textract-output/ t/	730 days
comprehend-out put/	730 days

4.4 Authentication — Cognito User Pool

Purpose

- Handle user authentication securely
- Avoid custom auth logic

Features

- Username: Email
- Email auto-verification
- OAuth2 Authorization Code flow
- PKCE enabled (SPA best practice)
- Hosted UI enabled

Why Hosted UI

- Faster MVP
 - Secure defaults
 - No password handling in frontend
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5. What Phase 1 Does NOT Include

Explicitly excluded (by design):

- ❌ Lambda functions
- ❌ Textract or Comprehend execution
- ❌ DynamoDB tables
- ❌ IAM roles for processing pipelines

These are handled in **Phase 2**.

6. Deployment Artifacts (Phase 1)

Included

- S3 automation scripts (buckets, encryption, lifecycle)
- CloudFront + OAC automation script
- Cognito User Pool CloudFormation template

Not included yet

- Lambda code
 - IAM roles for AI processing
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7. Security & Risk Considerations

Risks addressed

Risk	Mitigation
Public S3 access	Block Public Access + OAC
Credential leaks	Cognito Hosted UI
Data retention cost	Lifecycle rules
Bucket misuse	Separate buckets

8. Phase 2 Preview

Phase 2 will add:

- Lambda to submit Textract jobs
- Lambda to poll Textract results
- DynamoDB for job tracking
- Comprehend async jobs
- Vocabulary extraction logic

Phase 1 was designed **specifically** to support this without refactoring.

9. Phase 1 Status

- ✓ Architecture finalized
 - ✓ Security baseline applied
 - ✓ Website delivery operational
 - ✓ Authentication operational
 - ✓ Ready for AI pipeline integration
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10. Repository Structure (Recommended)

```
/docs
└── phase-1.md
/scripts
├── s3_setup.py
└── cloudfront_oac.py
/cloudformation
└── cognito-user-pool.yaml
```
